

Optimizing supply chain performance through sustainable manufacturing practices and capabilities in emerging markets: an integrated linear (PLS-SEM) and non-linear (fsQCA) approach

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ABSTRACT

Supply chain uncertainties are a key managerial concern because of their adverse influences on organizations and supply chain operations. Supply chain firms must establish a strategic construct: the capability to reconfigure, adapt, and respond during disruptive events. This study aims to empirically examine the impact of the dimensions of supply chain analytics capability and sustainable manufacturing practices, thereby achieving high levels of supply chain performance in emerging markets. The study combines the theoretical framework using dynamic capability theory to reconfigure supply chain firms' resources and capabilities to respond to disruptive events, thereby attaining a competitive advantage in dynamic markets. By employing the proposed questionnaires, participants are collected from 333 senior supply chain professionals involved in various manufacturing industries in Bangladesh. Accordingly, the study proposed five hypotheses to test the empirical validation using partial least squares structural equation modeling (PLS-SEM) and fuzzy-set Qualitative Comparative Analysis (fsQCA). The results show that management capability, data-driven culture, supply chain management, and sustainable manufacturing process significantly impact supply chain performance using the PLS-SEM model. In the fsQCA model, the findings reveal that management capability and supply chain management are sufficient core conditions to achieve high supply chain performance. This substantial result identifies that deploying dynamic capabilities techniques for the implication of extant capabilities in combination with processes for expanding and employing new insights strengthens the supply chain's capability to find potential uncertainties and respond to them rapidly.

1. Introduction

A supply chain (SC) contains environmental, product, and economic flows, and balancing these factors has a direct influence on a supply chain firm's comparative position (Ghobakhloo et al., 2023). Owing to the globalization of digital and production competencies, supply chains have become more effective in attaining sustainable comparative benefits (Wiredu et al., 2024). Consecutively, solutions for transforming and exploring information and resources have been involved in modern supply chain practices (Nandi et al., 2020). In this regard, consumers

face numerous disruptive events when realizing the value of the extant operations and the legality of the products and services in modern SC functions. Moreover, since paper-based functions exist in most supply chain firms, this minimizes the compatibility and transparency of stakeholders in supply chains (Budler et al., 2024). This disruptive event is more viable in mineral SCs due to the managerial and analytical complications of its levels from starting to ending periods (Nikookar et al., 2024). With the growing concern in demand for resources and capabilities in present decades, SCs have also confronted new disruptive events (Belhadi et al., 2024; Schleifenheimer & Ivanov, 2024).

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Accordingly, technical transformation, adverse social-economical barriers, and financial volatility implying sustainable development have severely infected production and management functions (Iftikhar et al., 2024). On the contrary, customers are finding that buying goods has fewer adverse effects on the environment and society (Holgado et al., 2024; Tosarkani et al., 2020).

When supply chain organizations face a turbulent environment brought about by the emergence of supply chain analytics, they choose capabilities and resources to balance this transforming function by combining and supporting dynamic capabilities that are accessible to them to employ in responding to disruptive events (Ahmadi et al., 2024; Chaudhuri et al., 2024). Dynamic capabilities provide supply chain firms with specific unique benefits and transformations as they cope with holistic change in achieving sustainable SC performance (Dubey et al., 2024). This also implies making decisions on achieving SC performance (SCP) and employing the combined changes. Thus, manufacturing firms employ innovative techniques and analytical capabilities to strengthen the sustainable performance of their supply chains (Budler et al., 2024). Supply chain analytics (SCA) may provide more than \$425B for the partners of the manufacturing sector by 2025. Such capabilities and resources lessen 610 M tonnes of CO₂ emissions to the enhancement of social and environmental aspects (World Economic Forum, 2017). However, SCAs can contribute a significant role in societal and environmental sustainability while assuring social and financial prospects (T. Wang et al., 2024). In this vein, the application of supply chain analytics facilitates sustainable performance and growth in supply chain networks in various industries (Fosso Wamba et al., 2020; H. Gupta et al., 2021). This kind of issue upholds a couple of validating research questions, which are the objective of this research.

RQ1. What are the dimensions of improving supply chain performance?

RQ2. How does supply chain performance play a role in supply chain analytics capability and sustainable manufacturing practices?

Urban shutdown, operational threats, and changes in workstations and dynamic market have speeded the function of supply chain analytics and sustainable manufacturing practices, interested concern from supply chain firms, and become one of the latest ideas in operation management (Frank et al., 2019; Holmström et al., 2019; Sawik, 2022). Emergent capabilities and resources, such as technical, management, data-driven capabilities, supply chain management, and sustainable manufacturing process, have accelerated the supply chain performance (Abdul-Rashid et al., 2017; Acquah et al., 2024; Munir et al., 2024). Sustainable SC is a concept that focuses on the dynamic application of data to improve operation and production functions, enhance SC transformation and upholding, and attain overall managerial and quality development (Abdul-Rashid et al., 2017). The utilization of supply chain analytics capabilities enhances the continual tracking of manufacturing goods and services, ensuring the higher quality of services and goods and lower operations expenditures in supply chain organizations (Pitelis et al., 2025). Despite extending the significance of dynamic capabilities in balancing complex supply chains, some technical, manufacturing, natural, and organizational obstacles have a deterrent impact on supply chain performance (Caiado et al., 2025). To enhance this scenario, recent studies in this domain have concentrated on addressing supply chain complexities (da Silva et al., 2025; Yousefi & Mohamadpour Tosarkani, 2022). Thus, it can be stated that most disruptive events arise from the shortage of specific concern about the significance of SC performance enablers and operational conservatism. Therefore, employing technical, management, data-driven capabilities, supply chain management, and sustainable manufacturing processes in the SC function from dynamic capability aspects can, directly and indirectly, affect the components of supply chain performance.

Our study adopted a multimethodological approach grounded in strong data gathered through survey questionnaires from more than thirty Bangladeshi manufacturing firms. This research makes threefold contributions. First, we suggest that by employing dynamic capabilities,

supply chain firms can combine supply chain analytics capabilities and sustainable manufacturing practices to achieve SC performance. We also reveal that this allows supply chain organizations to improve dynamic capabilities and as a part of the sustainable supply chain performance (Caiado et al., 2025). Second, we show that the validated hypotheses developed based on the dimensions, thereby accepting the proposed framework that enables SC firms to achieve SCP. Third, a multi-methodological approach has become a growing attention to employ research in operation management. We used linear (PLS-SEM) and non-linear (fsQCA) effects to measure the validated research framework.

The remainder of this research is developed in the following sections: Section 2 provides the concept development ideas and employs the theoretical framework. Section 3 shows the development of the proposition, and section 4 provides the overall framework of research methods, including data collection, context development, and measure validation. In addition, section 5 shows the data analysis and results, and section 6 shows the study's overall discussions, contributions, and implications.

2. Literature review

2.1. Dynamic capability view

Dynamic capability theory is embedded in the explanation and criticism of the resource-based theory (Barney, 1991). Researchers have suggested that a resource-based view (RBV) cannot be employed in dynamic environments where policy-makers in the markets are not perpetually visible, market environments are not understandable, and alters frequently occur (Eisenhardt & Martin, 2000). The dynamic capability theory suggests that supply chain firms evolve in dynamic competition to add new prospects and capabilities produced from incomparable, remarkable, and valuable resources (Teece et al., 1997). Accordingly, a supply chain firm is a dynamic function composed of capabilities, resources, and procedures to manage unforeseen crises (Gruchmann & Seuring, 2018). In response to rapidly turbulent environments, supply chain firms are required to enhance and employ a dynamic capability view (DCV) by adapting their systems and processes (Eisenhardt & Martin, 2000). As per the dynamic capability view, supply chain firms are suggested to employ higher-order resources and capabilities that further associate them with developing their response, reconfigure, and recovery capabilities to disruptive events (Chatterjee et al., 2024). As a result, supply chain firms can focus on responding to the changes in the environments, reconfiguring the resources and capacities to transform, and recovering the capabilities and resources to achieve comparative benefit during disruptive events (Bahrami & Shokouhyar, 2022; Dovbischuk, 2022; Teece, 2007).

DCV's aim of reconfiguring the internal and external capabilities and resources to manage external threats supplements the notion of developing resilience by rapidly recovering from turbulent environments (Bag et al., 2019; Irfan et al., 2022). The DCVs of a supply chain firm can enhance the degree and speed of the firm's recovery across any unforeseen circumstances (Chatterjee et al., 2024). Furthermore, higher-order resources and capabilities of dynamic capabilities like response, reconfigure, and recovery capabilities are presumed to be the adjustive functions of organizational resilience (Teece et al., 2016). The dynamic capability view provides supply chain firms with a theoretical roadmap for realizing and developing the necessary capabilities required for imposition (Chaudhuri et al., 2024). Caiado et al. (2025) also suggested that dynamic capability lays a theoretical framework for measuring how supply chain firms can establish the necessary capabilities to balance disruptive events efficiently. Moreover, DCV sheds light on dynamic environments, including technological innovation, crisis management, knowledge management, contingency planning, agile function, and resource reconfiguration to manage unforeseen events (AL-khatib & Ramayah, 2024). It spotlights the importance of adjusted capability, collaboration, resources, innovation, and continuous enhancements,

which broadly strengthen the improvement of resilience supply chains during disruptive events (Mora-Contreras et al., 2024). Thus, supply chain firms that employ DCV notions in their resilient techniques are better located to proactively manage, analyze, and lessen disruptive events, thus bolstering the performance of their supply chain organizations (Ruel & El Baz, 2023).

2.2. Supply chain analytics capability

DCs are referred to as top-level capabilities that balance resources and capabilities to achieve the SC performance of a firm in a dynamic environment (Teece, 2014). The founding blocks of the dynamic capability are necessary for SCA as they support digital innovation, data-driven culture, managerial technology, and resource management capabilities to enhance supply chain performance during disruptive events (Akter et al., 2016; Bahrami & Shokouhyar, 2022). Portraying the dynamic capability theory, SCA is the capability as a holistic analytic approach that facilitates rigorous knowledge for concurrent decision-making employing several human, data-driven, and technological capabilities in response to disruptive crises (Wong & Ngai, 2023). Such an analytics function employs location tracking, web-trim data, transaction data, visual data, and market data from internal and external sources to strengthen decision-making and insights (Wamba et al., 2017). For instance, Roßmann et al. (2018) revealed the application of SCA technology to increase demand forecasts, lessen inventory shocks, and enhance supply chain performance to achieve competitive advantage. Also, Zhao et al. (2017) demonstrate SCA capability as a beneficial technology for developing a streamlined framework for dynamic supply chain activity. Consecutively, researchers denote how it is desperate to merge SCA management capacity to enhance SC performance, establish settlement techniques in small and large-scale data breaches (Kude et al., 2017), and moderate the disruptive events of default of credit and transaction in the SC firms (Tsao, 2017).

Such a disruptive event in SCA enhances the capability to mediate and measure data at a minimum cost, extending enormous threats and opportunities for SC firms. SC firms that can integrate data from within and outside the firm and employ it efficiently achieve a comparative benefit (Li et al., 2025). Such capability building needs a necessary investment of capacities and resources to ensure competitive advantage and achieve sustainable SC performance during disruptive threats (Bak et al., 2023). It is obvious in the extant literature that applying SCA capability for the efficient utilization of data results in enhanced strategic and managerial decision-making, which, in turn, achieves better SC firm-level management and improves resilience during turbulent events (Jiang et al., 2024a; Matthias et al., 2017). People with these capabilities are less likely to withstand the application of unique knowledge management technologies, which improves the productivity of supply chain analytics composition (Munir et al., 2024). Employees also can identify suitable data to be examined to present significant insights for improving their decision-making (Wamba et al., 2017). However, SC organizations are interested in employing SCA capability to adjust to, reconfigure, and develop in changing markets (Bhatti et al., 2024).

2.3. Supply Chain Performance

The robust ecosystem needs dynamic innovation that is able to respond to global demand inclination and the necessities for sustainable business model development (Ngo et al., 2024). The implementation of sustainable growth is extensive in scope. However, overall, it ascertains the combination of financial growth, natural safeguard, and societal progress (Saha & Rathore, 2024), which facilitates supply chain organizations upright their numerous partner's needs (Bartolacci et al., 2020). Dynamic innovation, which can manage the requirement for economic and natural sustainability and the appeal of societal prosperity, plays a significant contribution in absolutely taking business and

managerial prospects into account during the digital innovation system (Lubberink et al., 2019), to meet the capability and resource requirements of present and future invention while not molesting the circumstances. However, dynamic innovation presents a new process for supply chain organizations to manage the natural, social, and financial prospects to become more resilient and sustainable during disruptive events (Wong et al., 2024).

Sustainable supply chain performance (SSCP) plays an important role in the dynamic processing of a supply chain organization. The timely and effective assessment of the SCs facilitates strategy function, process, and implementation of the business operations (Gawankar et al., 2017). Tracking the circumstances is becoming an important challenge with the growing dimension and comprehensive functions of the SCs. Consumers and buyers will recognize the real benefits of the services and goods if the SC provides transparency and presents features of the numerous traditional events connected with the goods and services (El Bhilat et al., 2024). SCs aim to connect partners' characteristics relating to economic and natural processes while attaining presumed SC performance and balancing raised ethical and operational standards (M. Kumar et al., 2024). Combining sustainability processes within SC organizations has become a top-notch priority in SC management design and operation (Kusi-Sarpong et al., 2019). The manufacturing and production processes are measured and decorated using sustainable capacities, such as waste management, reconfiguring resource usage efficiency, and carbon emissions footprints (Sarkis & Zhu, 2018).

Furthermore, in the extant literature, sustainable supply chain performance has been dominantly investigated in both managerial and academic insights. Scholars from many studies investigating SSCs have spotlighted the incorporation of financial, environmental, and social domains of sustainable performance (Belhadi et al., 2020; Formentini & Taticchi, 2016). Moreover, supply chain firms striving for sustainable operations are exposed to unforeseen alterations if they do not aim to balance the disruptive threats that lead to resilience in the competitive markets (M. Kumar et al., 2024). More insightful managers in such circumstances are more likely to support SC firms that do not recognize their dynamic market demands (Waqas et al., 2024). The adverse matter is when these dynamic market demands are transforming at changing with high speed and in unforeseen environments (Jia et al., 2024). Thus, SSCP and SCP investigate the sustainability facilitated by SC firms by the impacts of a global domain with its features of highly disruptive SC networks, global competition, and technological innovation (Hong et al., 2018). Therefore, the research framework is developed in Fig. 1 based on the current literature on supply chain performance in the context of disruptive events.

3. Hypotheses development

3.1. Technical capability

Technical capability pertains to understanding how to employ new competencies or technologies to obtain meaningful ideas from big data or large data sets (Parkhi et al., 2022). This is an essential capability required to ensure that technologies can cope with numerous information and data flows in any environment (Rialti et al., 2019). To employ resourceful data and information obtained from supply chain data, SC organizations must establish an environment comprising various datasets, devices, and functions (Jiang et al., 2024a). Accordingly, Gupta & George (2016) show that many systems include artificial intelligence, big data analytics, analytical analysis, data extraction, and curation. SC firms must obtain inserted technological networks that can manage various datasets simultaneously (Ghobakhloo et al., 2023). Technological innovation alone cannot cope with the adverse information and data of SC firms and requires broad support in professional workshops to achieve these managing skills during disruptive events (Wamba et al., 2017). It is obvious that scholars often employ technologies including machine learning, data curation, real-time decision-making, and

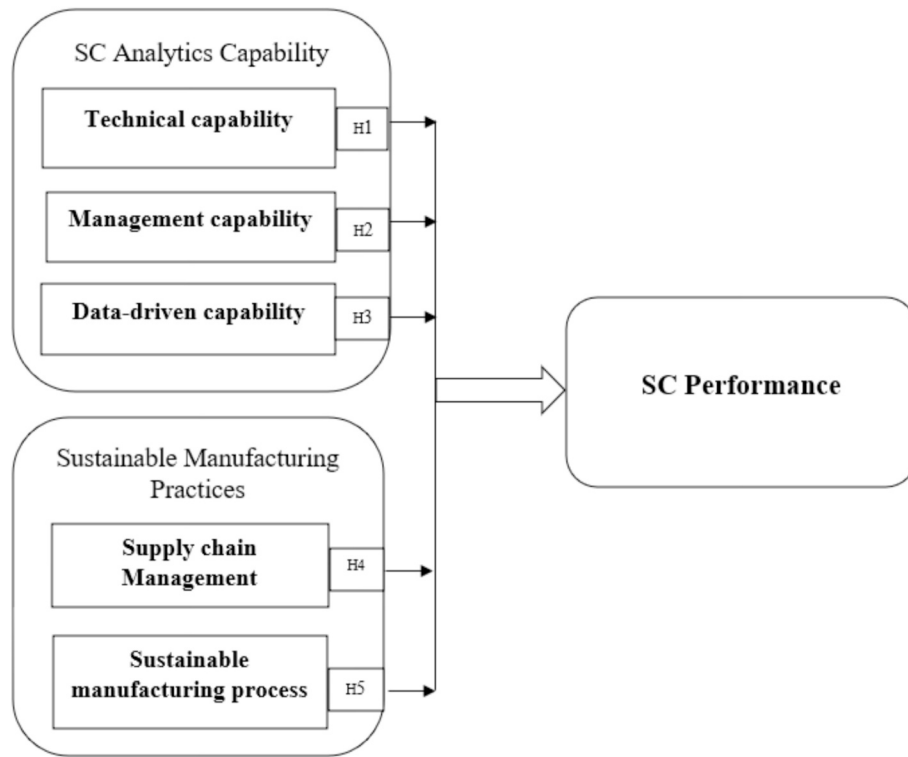


Fig. 1. Research framework.

predictive capabilities interchangeably (Jiang et al., 2024b; Srinivasan & Swink, 2018), to relate the supply chain capabilities that enhance SC organizations to gather stock and manage data to acquire useful ideas which can achieve comparative benefit (Gupta et al., 2020). Therefore, we formulated the following hypothesis:

H1: Technical capability positively affects supply chain performance, though this may vary with the level of technology integration.

3.2. Management capability

Management capability is the second significant antecedent for applying an assortment of applicable supply chain analytic structures and ascertaining what capability to determine from dynamic data (Gama & Magistretti, 2023). SC managers must acquire the necessary data discovery ability for data-driven decision-making (Ghobakhloo et al., 2023). People with this innovative capability are less likely to withstand the endorsement of new technologies of information systems, which allows the efficiency of the supply chain analytics structure (Sjödin et al., 2024). Employees also need the capability to determine the valuable data to be measured to outline practical implications for their decision-making (Wamba et al., 2017). Furthermore, management capabilities are often required through long years of working environment to achieve comparative benefit (Gama & Magistretti, 2023). Technical capability may be employed by innovating or growing in a dynamic market, whereas management capabilities are scarce to produce without appropriate resources and capabilities in supply chain firms (Parkhi et al., 2022). Thus, management capabilities are a significant source for analytics operations as policymakers plan a significant contribution in heading and culture-making roles (Xia et al., 2024). Therefore, we formulated the following hypothesis:

H2: Management capability is expected to enhance supply chain performance, depending on organizational adaptability.

3.3. Data-driven capability

A supply chain culture based on data-driven analytics is a highly

dynamic principle that needs to be realized and described. Over the past century, supply chain managers have recommended various definitions of data-driven culture, yet there is no solid account of a single notion. While many researchers recommend that a data-driven culture addresses nearly all sectors of a supply chain firm, others consider it a stick that holds a supply chain firm together to predict appropriate decision-making during disruptive events (Abubakar et al., 2024). Previous scholars in supply chains have considered data-driven culture as a way of sustained supply chain performance to achieve competitive advantage (Cristofaro et al., 2025; Zhang et al., 2024). Similarly, existing literature in big data analytics shows that a data-driven culture is significant for the performance of the supply chain's big data strategies. For example, Zhang et al. (2024) suggest that the causes of supply chain analytics' ineffectiveness relate to data-driven culture rather than to the features of data and the shortage of digital analytics. A supply chain in which the data-driven culture influences decisions is unlikely to achieve any return on its supply chain investments (Szukits & Móricz, 2024). Accordingly, the exertion to gather an enormous amount of SCA capability, attain technology, and make data-driven capability will be absorptive to manage the SCA in the decision-making process. Therefore, we formulated the following hypothesis:

H3: Data-driven capability positively influences supply chain performance, contingent on data governance maturity.

3.4. Sustainable supply chain management

Sustainable supply chain management (SSCM) is described as the management of the integration, finance, and materials evolved in supply chain firms, and as the incorporation and relationship among SC facilitators, including partners, customers, and suppliers, and the application of all the dynamic goals in the context of natural, social, and financial dimensions (Acquah et al., 2024; Tseng et al., 2021). Alkaraan et al. (2025) defined SSCM as a supply chain incorporative framework combining managerial activities and functions that effectively and efficiently manage service management, production process, and value delivery to fulfill partners' demands and enhance SC organization's

resilience and competitiveness in the long run. However, incorporating sustainability notions into the supply chain is significant for organizations to attain comparative benefits, especially in dynamic market circumstances (Anjomshoe et al., 2023). Additionally, sustainability has unenviable impacts on supply chain organization turnover, denoting a need to derive the productivity benefits of employing sustainability practices (Poursoltan et al., 2021; Tseng et al., 2022). Sustainability is growing more significant for supply chains due to the disruptive event developed by uncertainty, as unforeseen and unavoidable environments, such as financial and environmental threats, digital competencies, and political turmoil, happen more rapidly and are more adverse (Gölgeci & Kuivalainen, 2020). Therefore, we formulated the following hypothesis:

H4: Sustainable supply chain management contributes to performance, with effects influenced by stakeholder and regulatory contexts.

3.5. Sustainable manufacturing process

Manufacturing makes a significant contribution to every nation's economy, including supporting goods to serve the utilization required community and raw materials for transformation by other industries (Acquah et al., 2024; Kamal et al., 2020). Specified by the domination of the manufacturing sector, various nations may face financial crises, presuming there is a lack in their functions (Alkaraan et al., 2025). In this vein, the influences of the manufacturing process can be harmful, such that it influences natural and societal sustainability through random capabilities, resource production, and unfeasible manufacturing systems (Acquah et al., 2024). Sustainability SC interest require that manufacturing firms lessen the adverse effects of their systems on the economy, community, and environment (Opoku, 2025). Sustainable manufacturing activities should be facilitated in SC organizations to lessen resource and energy expenditure and the changing functions of waste developed from manufacturing functions (Afum et al., 2020). Bhatt et al. (2020) stated that sustainable manufacturing concentrated on lean and green notions rather than the complexity of sustainability. Holgado et al. (2020) stated the association between sustainable manufacturing practices and conservation activities with sustainable operational activities concerning financial, social, and natural benefits. However, supply chain firms evolved in manufacturing functions required to attain sustainability among economic, social, and environmental to sustain in the changing market, and sustainable manufacturing plays a significant role in carrying this sustainability in the long run (S. Liu et al., 2025). Therefore, we formulated the following hypothesis:

H5: Sustainable manufacturing improves supply chain performance, though cost and implementation complexity may affect this link.

4. Research methods

4.1. Context instruments

In order to examine the proposed paths of SCA capabilities and sustainable manufacturing practices, we employ the survey approach for empirical analysis after integrating the extant theories using a dynamic capability view to investigate the validated propositions. The survey approach uses the survey feedback on a five-score Likert scale, with "1" being "strongly disagree" and "5" being "strongly agree". Our survey approach focused on how SC manufacturing organizations in Bangladesh evolved in manufacturing products. Bangladesh is witnessing a developing country in the utilization of big data analytics solutions, and some supply chain managers are seriously investing in SCA capabilities to achieve a high level of sustainability in the manufacturing systems to attain a competitive advantage. It is reported that some Bangladeshi organizations are at several levels of employment of an SCA capability, which supports the Bangladesh domain appropriate for our research. This study aims to focus on small- and medium-sized supply chain organizations and choose the respondents from the manufacturing

industry in Bangladesh. This technique assures a well-rounded and targeted participant composition.

Our Study chooses the context of Bangladeshi manufacturing firms to choose the participant's industry. Bangladesh's logistics industry faces higher warehousing and transportation bottleneck costs because of poor networking structure and dependency on energy-oriented resources. The shortage of contemporary SC functions results in substantial waste, especially for decomposable products, with a 40 % consumption proportion influencing SC expenditures (The Business Standard, 2022). This raises economic recessions on contracts, eliminating efficiency in supply chain performance models. In the textile and garments section, the slowdown at Mongla and Chittagong, the major logistic transportation area in Bangladesh (BD), faces various bottlenecks, for instance, delayed shipments and high freight expenditures, thereby reducing contract performance (Infor, 2024). In this industry, computing for over 80 percent of BD's exports is notably susceptible to SC inefficiencies (International Monetary Fund, 2018). Excessive logistics expenditures, which accounted for 15–20 % of entire SC expenditures, reduce contribution margins in SC contracts.

The study shows that SCAC optimizes demand forecasting reliability, allowing manufacturing organizations to respond to consumer environmental awareness, in according with Zhu et al. (2007). Christopher & Ryals (2014) stated that SCAC ascertains demand prediction and responsiveness to environmental awareness (β)-oriented changes, enhancing SC performance metrics, for instance, logistics reliability and cost efficiency. SMP incorporates an influencing dimension that minimizes green expenditures and strengthens sustainability ratings, especially in high environmental awareness domains (Zhu et al., 2007).

4.2. Sample, participants, procedures

The survey aims to find manufacturing firms with a particular stage of SCA located in the Dhaka region of Bangladesh that broadly employ highly analytic projects in their business operations. The survey questionnaires were shared with the executive and senior management of supply chain firms to ensure that the collected data correctly showed the operating functions of the target SC surveyed firms. The survey questionnaires were shared using What's Up, email, and online and offline surveys to the respondents. WhatsApp and phone reminders were employed to increase the response rate of the survey questionnaires. In total, 348 full questionnaires were shared with the participants via email and offline, and 339 were answered accurately. Invalid survey respondents with many missing scores were excluded, and lastly, 333 credible questionnaires were obtained, with a response rate of almost 95 percent. Assuring the anonymity of the participants is a top priority for many reasons: it motivates honesty and unbiased samples, enhances higher respondent scores, and ensures ethical considerations by shielding the confidentiality and privacy of the respondents. The survey instruments were distributed on the purposive sampling approaches after assuring that all control constructs (invalidation of CMB) were selected to ensure the variance and accuracy of the measurement items. To find the possible biases based on common method bias, the survey questionnaires (a) the survey questions were taken from relevant literature, and (b) were carefully taken to safeguard the anonymity of the participants. (c) well taken in domains of proposed variables (MacKenzie & Podsakoff, 2012).

Most of the survey responses (333) are collected from medium-sized supply chain organizations as medium to larger firms usually have better access to capacities and resources and balance over activities reported in Table 1. Also, survey data is obtained from various industries related to manufacturing firms in Bangladesh, including automotive parts, food manufacturing, health, and medical instruments, pharmaceutical, textiles and apparel, and others manufacturing industries. In this research, participants including senior executives (50.5 %), manager (27.9 %), general managers (7.5 %), Chairman or CEO (3.9 %), and others (10.2 %), which means that participants were well adjusted to deciding how to

Table 1
Demographic Characteristics.

	Participant's (supplier's) Industry	No. of Participants
Participant's (supplier's) Industry	Automotive Parts	23
	Food Manufacturing	56
	Health and Medical Instruments	93
	Pharmaceutical manufacturing	69
	Textiles and Apparel Manufacturing	32
	Other manufacturing industries	60
	Total	333
	Senior Executive	168
Employee's Level	Manager	93
	General Manager	25
	CEO or Chairman	13
	Others	34
	Total	333
Number of Employees	Between 50 and 99	82
	Between 100 and 599	12
	Between 600 and 1,000	235
	More than 1,000	4
	Total	333

measure complex SC function issues of their firms and had a better insight of SC function and sustainable supply chain practices. Thus, the participants chosen were chosen based on the appropriate participants in the population.

4.3. Measure validation

We adapted the survey questionnaires based on the dimensions of supply chain analytics and sustainable manufacturing practices to analyze the proposed propositions employing five Likert scales: 5 (strongly agree) to 1 (strongly disagree). We adopted a total of thirty-two measurement items to test the proposed propositions. We adapted supply chain performance as a dependent variable (Paulraj, 2011). The measurement scales of supply chain performance include (SCP1), (SCP2), (SCP3), (SCP4), and (SCP5). We operationalized technical capability on the TCA1, TCA2, TCA3, TCA4, TCA5 survey items. In this research, we operationalized management capability (Dubey et al., 2019; Wamba et al., 2017) by employing measurement items. The measurement items of management capability were developed like MCA1, MCA2, MCA3, MCA4, and MCA5. Data-driven culture adapted from (Dubey et al., 2019; Yu et al., 2021). The measurement items of data-driven culture were identified as DDC1, DDC2, DDC3, DDC4, and DDC5. Similarly, the supply chain management construct was taken from (Abdul-Rashid et al., 2017). We operationalized SCM on the measurement scale of items SCM1, SCM2, SCM3, SCM4, and SCM5. Therefore, a sustainable manufacturing process was adapted from (Abdul-Rashid et al., 2017). The measurement items including SMP1, SMP1, SMP1, SMP1, and SMP1.

4.4. Analytical methods

The primary analytical approach of our study is Partial Least Squares Structural Equation Modeling (PLS-SEM) as a linear effect using SmartPLS 4 software. This analysis technique perfectly matches the objective of our research, which is to find the relationship between supply chain performance and our selected construct. PLS-SEM is a well-established and well-suited model for handling any data, whether it is normally or non-normally distributed (Y. Liu & Meng, 2024). This great advantage allows us to have a comprehensive understanding of overall supply chain performance and facilitates the exploration of relationships among various factors influencing this issue. It is also a well-known method for managing exploratory and complex research that involves

multiple underlying dimensions or constructs (Dash & Paul, 2021). Examining both measurement and structural aspects, it uncovers latent relationships between constructs and outcome variables, making an ideal choice for this study (El Bhilat et al., 2024).

Qualitative Comparative Analysis (QCA) has become a valuable tool for studying organizational configurations as it finds the principle where outcomes can arise from complex interactions between multiple conditions as a non-linear effect using fsQCA 4 software (Haefner et al., 2023; Misangyi et al., 2017). Each condition contributes uniquely to the overall effects of the outcome and facilitates the advantage of identifying meaningful and possible configurations (Fiss, 2011). This technique can also differentiate core and peripheral conditions within these configurations and offers raw coverage and consistency that aid in decision-making for overall outcomes (Leppänen et al., 2019). While QCA was a suitable analytical approach for this study, the variables involved are not inherently binary. Therefore, we employed fuzzy-set QCA (fsQCA) to effectively capture the detailed and overall nature of the data, ensuring a comprehensive analysis.

5. Major results and findings

5.1. PLS-SEM results

Our analysis of PLS-SEM proceeds through two sequential stages. In the first stage, we checked the scores of the measurement model to determine the reliability and validity of our selected constructs, including an examination of factor loadings, reliability measures, convergent and discriminant validity (Sarstedt et al., 2023). It is an essential procedure as it ensures that the measurement instruments accurately represent the selected constructs. In the second stage, we analyze the structural model, focusing on overall model fit and predictive capabilities assessed using goodness-of-fit measures and relevance metrics. Deriving from the structural model (Fig. 3), the path coefficients and their statistical significance are analyzed to reveal the strength and direction of the relationships between the latent constructs, providing insights into the hypothesized connections. After that, the model fit and PLS-predict are shown in Tables 5 and 6. The Standardized Root Mean Square Residual (SRMR), a goodness-of-fit indicator for the model, as shown in Table 5, yielded a value of 0.051. Lower SRMR values signify a better fit, which aligns with the objectives of our study (Richter et al., 2020). The PLS-predict procedure, as shown in Table 6, further validated the model's out-of-sample predictive power, showing lower RMSE and MAE values compared to the traditional LM procedure, indicating strong predictive performance (Hair et al., 2019; Shmueli et al., 2019).

Starting from Table 1, we get an overall demographic overview of participants in our study. Moving on, Tables 2 and 3 have the analysis of the measurement model. We verify the variable's reliability, internal consistency reliability, validity of convergent, and discriminant validity by the suggestion of Jr. et al., 2017. As per the suggestion, the score of composite reliability (CR) for each construct should exceed 0.7 to ensure the reliability of our measures. The value for average variance extracted (AVE) has to exceed 0.5 to ensure that the selected constructs of our study are capable of capturing sufficient variance from their indicators (Bagozzi & Yi, 2012). In line with the recommendation of Hair et al., 2019 and Fornell & Larcker 1981, the score for Cronbach's Alpha (CA) should be greater than 0.7. As presented in the Fig. 2, all constructs in this study meet this criterion, with several constructs surpassing the minimum threshold. This indicates robust internal consistency and reliability of the measurement model. To assess discriminant validity, we employed Fornell & Larcker (1981) Criterion in Table 3. As per the suggestion from Henseler et al. (2015) The values for each construct should be less than 0.85 to ensure that each construct is distinct from other constructs in our study. As evidenced in Table 3, all constructs in our study satisfy this requirement, thereby confirming their discriminant validity and ensuring the robustness of our measurement model.

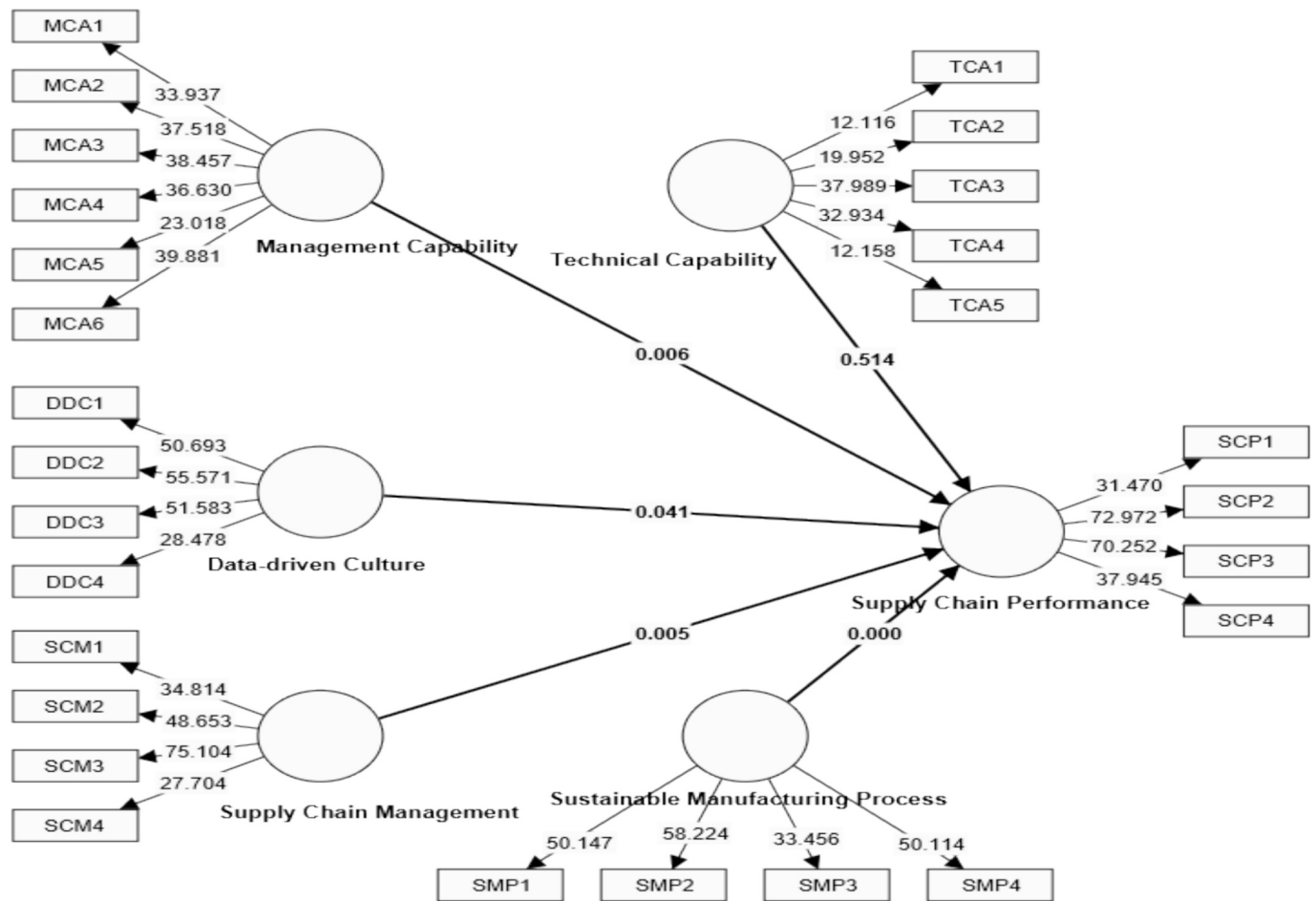


Fig. 3. Structural Model.

Moving to Table 4, we assessed the structural model procedure by examining the hypotheses outlined in our study. The score of the variance inflation factor (VIF) of each hypothesis measures the possibility of multicollinearity. All VIF scores were below the threshold of 5, confirming the lowest possible chance of multicollinearity (Jr. et al., 2017). Among the five hypotheses proposed in showing figure 3, four were supported based on a 5 % significance level ($p\text{-value} \leq 0.05$). The first hypothesis failed to reject the null hypothesis as the scored $p\text{-value}$ of 0.514 significantly exceeds the pre-determined $p\text{-value}$ of 0.05. However, the rest of the hypotheses are statistically significant as their $p\text{-value}$ is lower than 0.05, confirming the relationship with our selected variables. The second hypothesis strongly supports the claim of having a relationship between management capability and supply chain performance with a significantly lower $p\text{-value}$ of 0.006. Similarly, the third hypothesis, which proposes a connection between data-driven capability and supply chain performance, is supported, as evidenced by a $p\text{-value}$ of 0.041, lower than the 5 % significance level. Furthermore, the fourth hypothesis is also supported as the evaluated $p\text{-value}$ ($p\text{-value} = 0.005$) suggests a confirmed relationship between supply chain management. Lastly, the final hypothesis regarding the relation between sustainable manufacturing process and supply chain performance is strongly supported. The resulting $p\text{-value}$ of 0.000 is significantly lower than the 5 % significance level, providing robust evidence for this relationship.

5.2. fsQCA results

Qualitative Comparative Analysis (QCA) has become a valuable tool for studying organizational configurations as it finds the principle where

outcomes can arise from complex interactions between multiple conditions (Haefner et al., 2023; Misangyi et al., 2017). Each condition contributes uniquely to the overall effects of the outcome and facilitates the advantage of identifying meaningful and possible configurations (Fiss, 2011). This technique can also differentiate core and peripheral conditions within these configurations and offers raw coverage and consistency that aid in decision-making for overall outcomes (Leppänen et al., 2019). While QCA was a suitable analytical approach for this study, the variables involved are not inherently binary. Therefore, we employed fuzzy-set QCA (fsQCA) to effectively capture the detailed and overall nature of the data, ensuring a comprehensive analysis.

The calibration procedure is vital before starting the fsQCA Analysis. It involves transforming all variables into fuzzy set membership scores (Mello, 2021). As suggested by (Ragin, 2008a), we utilized the direct calibration approach to establish befitting crossover points. This will help to ensure that the resulting sets that will be generated through this analysis are meaningful and relevant for the decision to make. For this study, we are adopting 90 %, 50 %, and 10 % percentiles based on the recommendation from Pappas & Woodside, 2021. The 90th percentile represents the fully-in fuzzy membership score, while the 50th and 10th percentile represent cross-over and fully-out scores, respectively. The calibrated scores for all variables, along with summary statistics, are presented in Table 7.

After doing the calibration procedure, the next phase involves determining whether the causal conditions are necessary for the outcomes to occur (Schneider & Wagemann, 2010a). To call any of the six causal conditions of our study is necessary to affect the outcome, consistency scores for each condition should surpass 0.80 (Ragin, 2008b). Among them, a causal condition will be considered almost always

Table 2
Construct reliability and convergent validity results.

Likert scale constructs	Factor loading	Cronbach's Alpha (CA)	Composite reliability (CR)	Average variance extracted (AVE)
Technical Capability (TCA) (Dubey et al., 2019; Wamba et al., 2017)		0.796	0.86	0.555
TCA 2	0.767			
TCA 3	0.833			
TCA 4	0.799			
Management Capability (MCA) (Dubey et al., 2019; Wamba et al., 2017)		0.903	0.925	0.674
MCA 1	0.799			
MCA 2	0.824			
MCA 3	0.845			
MCA 4	0.818			
MCA 5	0.784			
MCA 6	0.853			
Data-driven capability (DDC) (Dubey et al., 2019; Yu et al., 2021)		0.885	0.921	0.745
DDC 1	0.873			
DDC 2	0.890			
DDC 3	0.868			
DDC 4	0.819			
Supply Chain Management (SCM) (Abdul-Rashid et al., 2017)		0.870	0.911	0.72
SCM 1	0.813			
SCM 2	0.862			
SCM 3	0.898			
SCM 4	0.817			
Sustainable Manufacturing Process (SMP) (Abdul-Rashid et al., 2017)		0.896	0.928	0.763
SMP 1	0.879			
SMP 2	0.883			
SMP 3	0.859			
SMP4	0.872			
Supply Chain Performance (SCP) (Paulraj, 2011)		0.909	0.936	0.786
SCP 1	0.858			
SCP 2	0.914			
SCP 3	0.908			
SCP 4	0.866			

Table 3
Discriminant validity.

Construct	1	2	3	4	5	6
Data-driven capability	0.863					
Management Capability	0.809	0.821				
Supply Chain Management	0.775	0.666	0.848			
Supply Chain Performance	0.742	0.744	0.694	0.886		
Sustainable Manufacturing Process	0.766	0.786	0.714	0.772	0.873	
Technical Capability	0.633	0.651	0.553	0.565	0.573	0.745

necessary if the score falls between 0.80 and 0.90 (Crespo et al., 2019; Ragin, 2009). If it surpasses the score of 0.90, then it will be called a necessary condition for our study (Rihoux & Ragin, 2009). From the scores of necessary conditions analysis shown in Table 8, we can conclude that none of the constructs have achieved the minimum threshold of 0.8 for representing high supply chain performance. However, two of the six constructs for low supply chain performance have achieved a score between 0.8 and 0.9, which can be considered almost always a necessary construct. The visualizations of the analysis of the necessary condition is shown in the Figs. 4 and 5.

Finally, Table 9 demonstrates the sufficient configurations that guided us to both high and low SCP, thereby Fig. 6 shows the trend of consistency and coverage values. First, we will discuss the combination set that can improve the supply chain's performance. Five models are generated from fsQCA sufficiency analysis, and almost all of them have the potential to contribute to high SCP based on the score of raw coverage. The overall solution coverage is 0.949, which significantly exceeds the threshold of 0.75 recommended by Schneider & Wagemann, (2010), indicating the robust explanatory power of our model. Among the five models, the 4th model has the highest raw coverage value that can simulate the real-life use case scenario with a satisfying consistency score (Ragin, 2006). This indicates that the combination of MCA and SCM has a 65.1 % chance of bringing out the high supply chain's performance. The same goes for the 3rd model, where the involvement of MCA and DDC has been spotted. However, the possible chance has come down to 63.2 %, which can be negligible. The 2nd model demonstrated a different approach with the involvement of DDC and TCA with a chance of 58.2 % chance to simulate. Model 1, with a 40.5 % chance, also shows a satisfactory performance compared to the least effective model, Model 5. Model 5 has scored the lowest raw coverage value, providing a poor solution to uplift the supply chain performance.

Table 9 also demonstrates the solution models that can reduce supply chain performance. Among the three models, we should be particularly concerned about the 6th and 8th models with a great raw coverage score. Model 8 has the best possible raw coverage score with an 87.6 % chance of decreasing supply chain performance. his model indicates that the absence of SMP can decrease the supply chain performance. The 6th solution model highlights the importance of not missing out on MCA, DDC, and SCM, as there is a 68.3 % chance of decreasing supply chain performance. The 7th solution model warns us that the absence of MCA, combined with the presence of TCA and DDC, has a chance of 34.49 % decreasing supply chain performance.

6. Discussion

6.1. Theoretical implications

This study is one of the earliest attempts to rigorously investigate the significance of the SC analytical capability and sustainable manufacturing practices to achieve supply chain sustainability in a supply chain firm from the perspective of supply chain performance. This research illustrates the potential of employing a dynamic capability view to address how SCP approaches, SCA capability and SMP can be deployed at various domains in the SCP to overcome turbulent environments. This research shows the empirical analysis of the antecedents that contribute to achieving supply chain performance during disruptive events. Through addressing how supply chain firms can strengthen their performance when dealing with disruptive events. This empirical validation connects to the theoretical insights of supply chain management by confirming its determining antecedents. This research connects investigation and utilization to achieve supply chain performance, revealing that for the supply chain to adjust and respond to unforeseen events in the firm, it is significant to integrate dynamic capabilities and sustainable manufacturing practices with new insights, exciting capabilities, and resources. This insight is about the extant study's conclusions that show the advantages of SCA capability (Kumar & Raj, 2024)

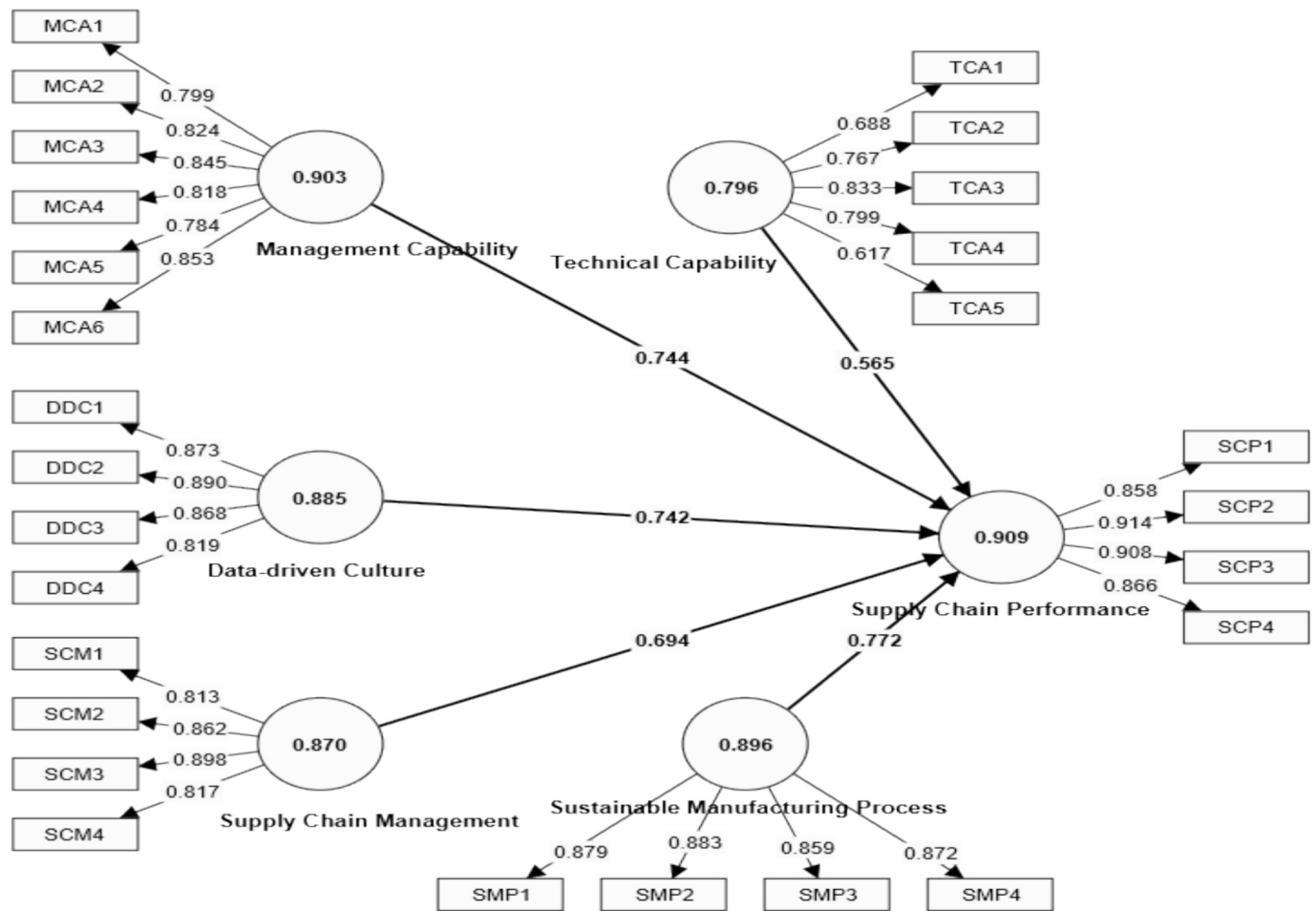


Fig. 2. Measurement Model.

Table 4
Hypotheses and Path Coefficient.

Hypothesis	Path	Beta (β)	T Statistics	P values	Decision	f^2	VIF
H1	Technical Capability $\rightarrow$ Supply Chain Performance	0.042	0.652	0.514	Not Supported	0.003	1.859
H2	Management Capability $\rightarrow$ Supply Chain Performance	0.21	2.766	0.006	Supported	0.035	3.851
H3	Data-driven capability $\rightarrow$ Supply Chain Performance	0.149	2.048	0.041	Supported	0.016	4.294
H4	Supply Chain Management $\rightarrow$ Supply Chain Performance	0.165	2.795	0.005	Supported	0.03	2.772
H5	Sustainable Manufacturing Process $\rightarrow$ Supply Chain Performance	0.351	4.682	0.000	Supported	0.117	3.261
Construct				R^2		R^2 Adjusted	
Supply Chain Performance				0.676		0.671	

Table 5
Model fit.

	Saturated model	Estimated model
SRMR	0.051	0.051
d_ULS	0.97	0.97
d_G	0.541	0.541
Chi-square	1027.364	1027.364
NFI	0.853	0.853

Table 6
PLS-predict.

Items	Q^2 predict	PLS-SEM_RMSE	PLS-SEM_MAE	LM_RMSE	LM_MAE
SCP 1	0.573	0.573	0.573	0.573	0.573
SCP 2	0.523	0.523	0.523	0.523	0.523
SCP 3	0.495	0.495	0.495	0.495	0.495
SCP 4	0.471	0.471	0.471	0.471	0.471

Table 7
Calibrations and descriptive statistics.

Configurational Constructs	Fuzzy Set Calibration					Min	Max	N Cases
	Fully-in	Cross-over	Fully-out	Mean	S. D			
Technical Capability	4.20	3.60	2.00	0.491	0.319	1.000	5.000	333
Management Capability	4.33	3.84	2.00	0.500	0.312	1.000	5.000	333
Data-driven capability	4.25	3.75	2.00	0.467	0.330	1.000	5.000	333
Supply Chain Management	4.25	3.50	2.00	0.541	0.328	1.000	5.000	333
Sustainable Manufacturing Process	4.25	3.75	2.00	0.451	0.337	1.000	5.000	333
Supply Chain Performance	4.25	4.00	2.00	0.567	0.284	1.000	5.000	333

Table 8
Analysis of Necessary Conditions for predicting Supply Chain Performance.

Configurational Constructs	High Supply Chain Performance		~Low Supply Chain Performance	
	Consistency	Coverage	Consistency	Coverage
Technical Capability	0.705	0.814	0.788	0.670
~ Technical Capability	0.573	0.639	0.576	0.507
Management Capability	0.763	0.865	0.844	0.731
~ Management Capability	0.562	0.639	0.583	0.504
Data-driven capability	0.715	0.868	0.857	0.696
~ Data-driven capability	0.568	0.606	0.515	0.476
Supply Chain Management	0.782	0.820	0.775	0.730
~ Supply Chain Management	0.492	0.608	0.585	0.467
Sustainable Manufacturing Process	0.702	0.882	0.876	0.691
~ Sustainable Manufacturing Process	0.559	0.579	0.467	0.447

and sustainable manufacturing process (Abdul-Rashid et al., 2017; Acquah et al., 2024). Another implication of this study shows that the theoretical framework on dynamic capability has a potential influence on supply chain performance, following the recommendation reported by Munir et al. (2024) regarding the significance of SCA capability and sustainable manufacturing practices to achieve supply chain performance.

This research examines how the dimensions of DCV contribute to the significance of supply chain performance during disruptive events. It addresses how SCA capabilities strengthen SC firms to respond to turbulent environments by building the most of their extant capabilities and resources while also subjecting them to new environments. This contribution of SCA capability within the perspective ventures a theoretical understanding of how DCV capabilities can develop an SC firm's capability to manage turbulent environments efficiently. Sustainable manufacturing practices (SMP) enable SC firms to effectively manage their operations and activities in the context of disruptive events. SMP optimizes green management techniques that benefit logistics and manufacturing organizations; thereby, their sustainable practices are integrated to offer green concerns in forward and reverse logistics. This research constructs on all-embedding dimensions to focus on sustainable manufacturing practices to achieve a sustainable comparative advantage in a robust supply chain. This theoretical framework reveals a dynamic approach for SC firm to strengthen their capabilities for SCP, thereby developing the theoretical foundation of SCM. In this study, SCP is measured by developing a wide range of antecedents that address all levels of disruptive events: technical capability, management capability, data-driven capability, supply chain management, and sustainable manufacturing process. Embedding these levels suggests a divergent perspective on SCP capabilities, understanding the significance of both the dimensions of SCA capability and SMP. This theoretical lens amplifies the comprehension of how SC firms can minimize the disruptive event's impact of interruptions across their supply chain. In recent times, SC firms have continuously subsidized and enhanced SMP expertise and structure, which improves insight into how this dynamic

capability connects with their capabilities and resources in the supply chain. The value-added supply chain analytics capability has a significant role in the connection between SMP and SOP because dynamic capabilities associate all supply chain stakeholders to manage numerous demands and associate all the distributors and vendors to meet their potential capabilities with the demands of the SC firm.

6.2. Managerial implications

Consistent with theoretical implications, the study shows various valuable managerial knowledge for supply chain organizations. The implications also facilitate practical contributions for SC policy-makers of manufacturing organizations who seek supply chain performance outcomes for their firms through sufficient solutions from SCP. The five recipes for high levels and three for low levels of SCP require diverging degrees; thus, not all recipes should be sufficient to attain a high level of SCP. Thus, when opportunities are focused, and the SC firm looks to develop them, it claims that there are already current manufacturing methods to attain competitive advantage during disruptive events. The SC managers must find critical paths in their manufacturing framework and reconfigure the necessary capabilities and activities in the SC firm.

Furthermore, the DCV theory reveals that SC firms contain dynamic resources that would assist them remain competitive in their manufacturing section in the long run. The sufficient configurations associated with SCP claim that the SC manager is devoted to employing the SCA capabilities and sustainable manufacturing practices for a high level of SCP. On the contrary, sufficient configurations for low levels of SCP claim that manager should not undermine achieving SCP because their negation or presence at low levels is broadly averse to the SC organization's aims. In this vein, SC policy-makers in manufacturing firms should invest profoundly in achieving SCA capabilities and sustainable manufacturing practices to attain their supply chain performance aims and goals. In addition, our results also produce five sufficient configurations for the high levels of SCP. For all these causal conditions, technical capability, management capability, data-driven capability, supply chain management, and sustainable manufacturing process investing in attaining comparative benefits have enhanced SCP during disruptive events. Thus, SC manufacturing firms should achieve these practices and capabilities and integrate them with MCA, SCM, DDC, and TCA to achieve SCP. However, the empirical validation of this research suggests SC policy-makers with a comprehensive formwork on SCP strategies. It strengthens the importance of SCA capabilities and sustainable manufacturing practices, enabling SC firms to overcome the disruptive events of uncertainties effectively. The managerial contribution shows that determining antecedents to SCP can facilitate SC firms in the early exploration and exploitation of possible uncertainties, thereby enabling operational sustainability.

The potential results of this study aligned with the practical contribution in other developed and developing countries. Likewise, our findings could be optimized to investigate emerging business challenges, such as backward SC infrastructure or government regulatory volatility. SC manufacturing practices can be adopted in local domains, empowering organizations to tailor cost-effective environmental technologies (Pagell, 2004). SCA capabilities, such as technical, management, and

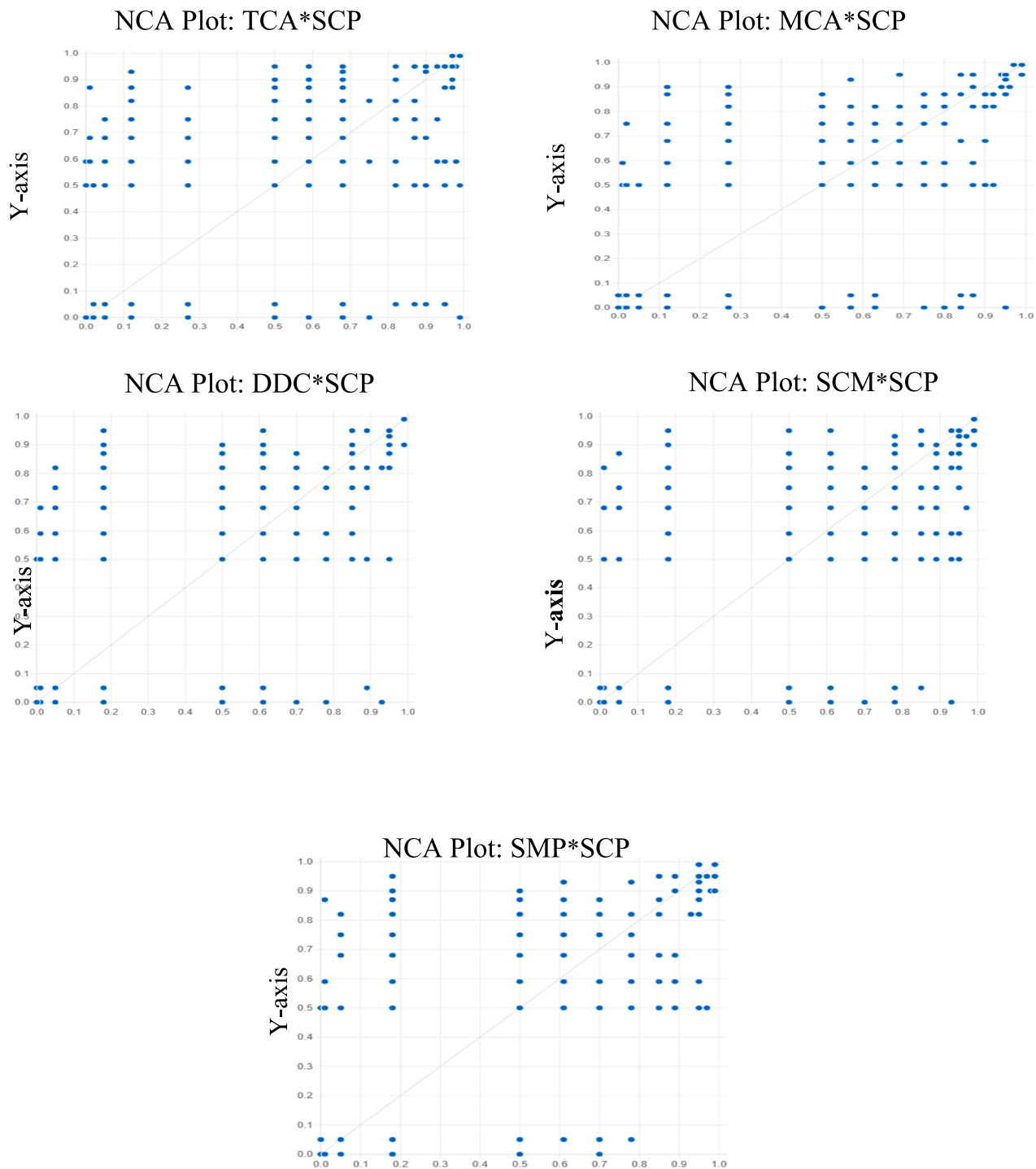


Fig. 4. NCA results for high level of SCP by fsQCA.

data-driven capabilities, can investigate market variability driven by customer trends or trading systems. SMP can enable manufacturing firms to observe rigorous green regulations in the US market while eliminating functional expenditures through energy-efficient policies (Seuring & Müller, 2008).

Our research also represents the significance of optimizing various manufacturing organizational networks, including transportation efficiency, carbon emissions, and logistics management, to eliminate expenditures and green influence. Reverse logistics, which entails controlling the outcome, reuse, or dumping of products and goods, directly facilitates optimizing efficiency by strengthening routing and

return shipments (Ambekar et al., 2022; Fani et al., 2025). This study will help managers optimize SCM and SMP, as well as manage sustainability, circular SC, and logistics strategies that frequently hinder successful SCP in manufacturing organizations (Sonar et al., 2022). Furthermore, the study presented how SMP, for instance lean production, supplement logistics techniques like environmental goods transport, strengthening a more coordinated SC sustainability framework. Likewise, this study illustrates on how manufacturing firms can leverage DCVs to adopt technical and managerial capabilities, for instance IoT-allowed firms tracking functions or data-driven logistics framework to strengthen SC efficiency and visibility in emerging markets. The study



Fig. 5. NCA results for low level of SCP by fsQCA.

also contributes to the circular supply chains by optimizing the tittle bottom line strategies using reuse, regenerate, and transform into new products or innovative products by eliminating emissions and waste (Khan et al., 2021). This research contributes particularly to supply optimization, which reduces costs while eliminating the manufacturing firm's environmental impact. Inventory optimization helps manufacturing firms employ such techniques, including eliminating additional stock, relevant waste and emissions, and demand-oriented inventory (Dehaybe et al., 2024). This implies that SCAC contributes to such techniques using data-driven supports and sustainable SC

processes, thereby effectively influencing green and sustainable SC performance (Wang & Dai, 2018).

Furthermore, the findings suggest that cost-maximizing contracts and sustainable manufacturing capabilities and practices in predicting SC performance, thereby considering modeling carbon emissions and trading by the government intervention in manufacturing firms (Gopal et al., 2024; Rathnasiri et al., 2022). It is a mutual obligation of manufacturing firms, governments, and NGOs to identify knowledge and enhance awareness about cost-sharing contracts along with carbon credits, emissions, and tax debates. However, the government should

Table 9
Configurations for achieving high and low levels of Supply Chain Performance.

Category	Model	Configuration	Raw Coverage	Unique Coverage	Consistency
High Scores for Supply Chain Performance	Model 1	SCM*~SMP	0.405	0.027	0.788
	Model 2	TCA*DDC	0.582	0.009	0.907
	Model 3	MCA*DDC	0.632	0.007	0.908
	Model 4	MCA*SCM	0.651	0.021	0.908
	Model 5	~MCA*~DDC*~SCM*SMP	0.268	0.007	0.884
Low Scores for Supply Chain Performance	Solution Coverage:	0.903		Solution Consistency:	0.800
	Model 6	~MCA*~DDC*~SCM	0.683	0.010	0.850
	Model 7	TCA*~MCA*DDC	0.344	0.003	0.840
	Model 8	~SMP	0.876	0.085	0.691
	Solution Coverage:	0.949		Solution Consistency:	0.643

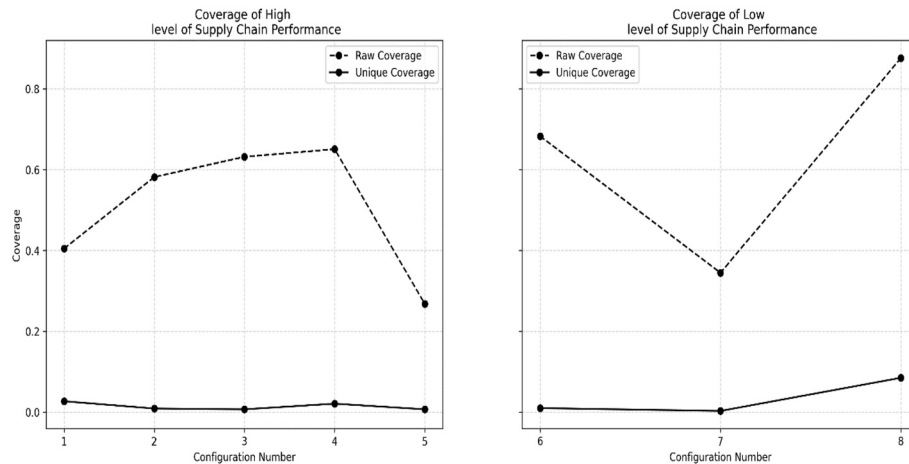


Fig. 6. The configurational model using coverage and consistency values.

force strategies and systems to the government body about green development programs. Likewise, policymakers can employ green management practices such as carbon credits for improving DC efficiency, emission trading for supporting circular SC, and tax rebates for supporting sustainability-driven strategies, thereby achieving high SC performance. These government systems coordinate with reverse supply chains, circular SC, environmental policy, and policy advocacy while eliminating regulatory crises.

6.3. Limitations and future research

This research's limitations could also be considered for future directions in this domain. This research develops the conceptual and theoretical validation foundation of the associations between SCA capabilities and sustainable manufacturing practices to achieve supply chain performance. It examines an empirical analysis through multi-methodological methods using PLS-SEM and fsQCA. Future research can use case-based analysis to further identify and measure this association from the perspective of the longitudinal supply chain organization process. Furthermore, our study primarily accumulates data from the supply chain manufacturing firms of Bangladesh, which is one of the core limitations of this research. Numerous natural, financial, and cultural constructs may lessen the paradigmatic and universality of the empirical validations. Future research may investigate these associations by gathering data from different sectors, states, and countries. In addition, future studies could use cross-sectional research models to use panel data to analyze the sequential effects of latent indicators. Also, future research should employ different dimensions of the constructs, for instance logistics and transportation that lead to sustainable supply chain performance can be achieved. Likewise, future research could employ SEM with a demand function to measure mediating effects to test sensitivity analysis considering the dimensions of environmental

awareness (β) or demand elasticity to predict high SCP. Therefore, the research nevertheless delves significant ideas and can be employed as a conceptual development for future research perspectives into the contribution that SCA capability and sustainable manufacturing practices play in facilitating supply chain firms to achieve high levels of SC performance.

6.4. Conclusions

This study is regarded as groundbreaking as it expands the influence of the supply chain performance of SC manufacturing organizations using PLS-SEM and fsQCA models. The theoretical foundation of this research contains important contributions to the recent literature, not just on the application of SCA capability but also on the implications of sustainable manufacturing practices to achieve a high level of supply chain performance. Accordingly, this research combines both dimension antecedents of SCP. It develops the theoretical context of "TCA $\rightarrow$ MCA $\rightarrow$ DDC $\rightarrow$ SCM $\rightarrow$ SMP" by integrating a dynamic capability view to test the proposed hypotheses using PLS-SEM and fsQCA models. In PLS-SEM results, the findings show that all the proposed hypotheses are supported except H1 to achieve SCP outcome. Accordingly, the findings from the fsQCA result show that five sufficient configurations are necessary to achieve high levels of supply chain performance. Management capability and supply chain management are significant antecedents as a core condition for high levels of SCP. Additionally, three sufficient configurations are produced in fsQCA models to achieve high levels of supply chain performance. The findings show theoretical and practical contributions for SC manufacturing firms to achieve supply chain performance capability, particularly step-by-step, using SCA capability and sustainable manufacturing practices to assist them in responding rapidly to disruptive events.

In summary, the outcome of this empirical result shows some

significant and interesting implications for the theoretical and practical discussions encompassing DCV theory, SC analytics capability, and sustainable manufacturing practices under the outcome variable of SC performance. This research investigated the influence of SCAC and SMP in controlling the effects of circular supply chains and predicting supply chain performance. First, the results illustrate that there is a direct and core relationship between the dimension of SCAC and SCP, except for the technical capability of SCP in PLS-SEM analysis. This implies that management capability and data-driven culture are more significant in predicting SCP. Accordingly, professionals should possess enough data analytics competence for better quantitative decision-making. Skillful manufacturing firms that seek to employ new SC functions and reduce operational costs and dependencies may utilize the knowledge retained from the data-driven capability to predict circular SC and enhance product delivery and quality. Regarding SMP, the results indicate an obvious connection with SCP. This is due to combining sustainability-driven considerations into the reverse logistics, transportation, and packaging of manufacturing goods to strengthen sustainable performance, including social, economic, and environmental gains for manufacturing firms.

Furthermore, this study offers insights into how the limitations of PLS-SEM can be ignored by combining its empirical analysis with the fsQCA method. The PLS-SEM method derives the linear effects of the proposed relationship that lead to a single or complex relationship, whereas fsQCA offers the combining effects of the non-linear relationship of the proposed framework that encompasses based on the all-causal conditions to find multiple combinations to predict a particular outcome. This methodological robustness offers novel insights into combining the effects of SCAC and SMP are produce diverse empirical configurations to predict SCP.

CRediT authorship contribution statement

Jewel Rana: Writing – original draft, Validation, Resources, Investigation, Conceptualization. **Md Zubair Ahmad:** Visualization, Software, Resources, Methodology, Investigation, Formal analysis. **Mohammad Fakhurul Islam:** Writing – review & editing, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Md Nazmul Islam Jihad:** Writing – review & editing, Visualization, Validation, Resources, Investigation, Formal analysis. **Md. Rashed:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Mohammed Kamruzzaman:** Writing – review & editing, Validation, Resources, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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